

VL-CLIP: Enhancing Multimodal Recommendations via Visual Grounding and LLM-Augmented CLIP Embeddings

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Abstract

Multimodal learning plays a critical role in e-commerce recommendation platforms today, enabling accurate recommendations and product understanding. However, existing vision-language models, such as CLIP, face key challenges in e-commerce recommendation systems: 1) Weak object-level alignment, where global image embeddings fail to capture fine-grained product attributes, leading to suboptimal retrieval performance; 2) Ambiguous textual representations, where product descriptions often lack contextual clarity, affecting cross-modal matching; and 3) Domain mismatch, as generic vision-language models may not generalize well to e-commerce-specific data. To address these limitations, we propose a framework, VL-CLIP, that enhances CLIP embeddings by integrating Visual Grounding for fine-grained visual understanding and an LLM-based agent for generating enriched text embeddings. Visual Grounding refines image representations by localizing key products, while the LLM agent enhances textual features by disambiguating product descriptions. Our approach significantly improves retrieval accuracy, multimodal retrieval effectiveness, and recommendation quality across tens of millions of items on one of the largest e-commerce platforms in the U.S., increasing CTR by 18.6%, ATC by 15.5%, and GMV by 4.0%. Additional experimental results show that our framework outperforms vision-language models, including CLIP, FashionCLIP, and GCL, in both precision and semantic alignment, demonstrating the potential of combining object-aware visual grounding and LLM-enhanced text representation for robust multimodal recommendations.

CCS Concepts

• **Information systems** → **Recommender systems**; **Users and interactive retrieval**.

Keywords

Multimodal Learning, E-Commerce, CLIP, Visual Grounding, Large Language Models, Image-Text Representation, Retrieval, AI for Recommendation

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1 Introduction

E-commerce platforms have revolutionized the way consumers interact with products, offering extensive catalogs that cater to diverse preferences. As the number of products continues to grow exponentially, delivering highly relevant personalized recommendations has become an increasingly complex challenge. Consumers often rely on multimodal interactions—searching with a combination of textual queries and images—to find the products they desire. Therefore, improving multimodal representation learning is critical for enhancing search accuracy, recommendation quality, and overall user experience in e-commerce [34].

Recent advances in vision-language models have significantly improved cross-modal retrieval. CLIP [23], in particular, has demonstrated strong zero-shot capabilities by aligning images and text in a shared embedding space. However, despite its success, CLIP exhibits several limitations when applied to e-commerce scenarios.

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Figure 1: Illustration of: (a) visual recommendation improvement achieved by our proposed model, VL-CLIP and (b) visual search improvement using VL-CLIP.

First, CLIP processes images globally, meaning that it often fails to capture fine-grained product attributes that are crucial to distinguish visually similar but semantically different items. For example, two handbags might appear nearly identical in a global embedding space, even if one has a unique texture or clasp design that differentiates it. This weak object-level alignment leads to suboptimal retrieval performance, especially in a large e-commerce platform.

Another major challenge is the ambiguity of textual representations. Product descriptions in e-commerce catalogs vary widely in quality and consistency. Some descriptions are too verbose, containing extraneous information, while others are sparse, lacking essential details. CLIP's text encoder struggles with such inconsistencies, especially with long-text descriptions, leading to poor semantic alignment between textual and visual representations. Without structured and enriched textual inputs, CLIP may misinterpret product intent, reducing the accuracy of multimodal retrieval.

Moreover, existing multimodal models are typically trained on general-purpose datasets, such as LAION-400M [25], which contain a broad spectrum of image-text pairs. While this training paradigm enables broad zero-shot learning, it also introduces a significant domain mismatch when applied to e-commerce. Product images often contain controlled backgrounds, well-lit professional shots, or lifestyle depictions, all of which differ from the diverse, noisy images seen in open-domain datasets. Consequently, pre-trained models fail to generalize effectively to e-commerce-specific data, necessitating domain adaptation strategies [14].

To overcome these limitations, we propose a novel framework that enhances CLIP embeddings through two key innovations: (1) the integration of **Visual Grounding** for fine-grained object localization and (2) the use of a **Large Language Model (LLM)** to refine textual embeddings. Visual Grounding [15] enables precise localization of key product attributes within an image, ensuring that CLIP's vision encoder focuses on the most relevant regions. By incorporating Visual Grounding, we improve object-level alignment, leading to more discriminative visual embeddings.

On the textual side, we employ an LLM agent to enrich product descriptions by generating structured, semantically meaningful text representations. Given raw metadata, the LLM refines descriptions, removes noise, and injects domain-specific knowledge, ultimately improving the quality of text embeddings. This augmentation mitigates CLIP's struggle with ambiguous text and ensures that the image-text alignment is robust, accurate, and context-aware.

Figure 1 illustrates the effectiveness of our approach in both visual and textual recommendation. In Figure 1 (a), the traditional recommendation system suggests products based on broad categorical similarity, often missing fine-grained visual coherence. In contrast, our visual recommendation system, powered by Visual Grounding and enhanced CLIP embeddings, retrieves visually and semantically aligned items, improving recommendation relevance. Similarly, Figure 1 (b) highlights how our model enhances e-commerce search. Traditional keyword-based search may yield inconsistent results when dealing with complex queries such as "*area rug with pet pic*" or "*damask silk bed sheet.*" Our model effectively aligns textual queries with the most relevant visual content, ensuring that search results are not only textually but also visually accurate. These improvements validate our approach's superiority in capturing fine-grained details and providing semantically meaningful retrievals, ultimately enhancing the user experience.

The contributions of our paper are threefold: First, we introduce a novel multimodal pipeline that integrates Visual Grounding and LLM-enhanced embeddings to improve fine-grained alignment in e-commerce applications; Second, we develop a scalable retrieval and ranking system that efficiently handles large-scale product catalogs; Third, we validate our approach through extensive experiments over tens of millions of items in *Walmart.com*, demonstrating significant improvements in retrieval accuracy, recommendation quality, and overall system performance compared to existing state-of-the-art multimodal models.

The remainder of this paper is organized as follows. Section 2 discusses related work in multimodal learning, vision-language

models, and e-commerce recommendation systems. Section 3 describes our proposed framework, detailing the enhancements to both image and text representations. Section 4 presents experimental results, including comparative evaluations and ablation studies. Section 5 concludes the paper.

2 Related Work

Multi-Modality learning has long been an active area of research. The advances in pre-trained vision language models enable applications across diverse domains such as healthcare [9, 20], finance [7], social networks [1, 22], search engines [6, 31], and e-commerce [10, 17]. Transformer-based architectures revolutionized multi-modal learning. By integrating textual and visual input into a unified latent space through self-attention and cross-attention mechanisms, models such as VL-BERT [26], ViLBERT [16], and LXMERT [27] laid the foundation for robust vision language reasoning. Subsequent models, including VisualBERT [11], UNITER [3], and OSCAR [13], further refined these capabilities, achieving state-of-the-art performance across multiple benchmarks and enabling generalized representation learning.

In parallel to attention-based mechanisms, Radford et al. introduced the CLIP [23] model, a dual encoder approach, trained on vast amounts of noisy image-text data. It sparked significant interest by showcasing robust performance across various vision-language tasks. Using contrastive learning mechanism to directly align visual and textual embeddings in shared space, it enabled impressive zero shot retrieval capabilities. Many works have extended CLIP by scaling up data [4], improving data curation [4, 24], altering inputs [8, 28], refining the loss function or alignment strategy [18, 29], adapting to new tasks [21, 32], ranking [33] and domain adaptation [5, 12].

Building on the capabilities of CLIP, we fine-tune its dual encoder architecture to adapt to the e-commerce domain, where multi-modal retrieval is critical for matching textual queries to product images. Our approach involves leveraging domain-specific datasets comprising noisy and diverse image-text pairs, a hallmark of e-commerce platforms. By tailoring CLIP to handle e-commerce-specific challenges, we aim to achieve superior alignment and retrieval performance, ultimately improving customer experience in search and recommendation systems.

3 Methodology

In this section, we introduce VL-CLIP, a systematic framework for fine-tuning the CLIP model to achieve robust image-text alignment in e-commerce applications (see Figure 2). The framework integrates advanced vision-language techniques across three stages: 1) image region refinement with Visual Grounding, 2) LLM-driven textual query synthesis, and 3) contrastive training with CLIP optimizations. Below, we provide a comprehensive breakdown of each component, including implementation specifics and design rationale. This robust approach addresses challenges of data noise, domain-specific alignment, and scalability. All the mathematical symbols used in this paper are listed in the table 8 in Appendix A.

3.1 Image Region Refinement with Visual Grounding

To focus on product-relevant regions, we employed Grounding DINO (GD)—a zero-shot object detection model that aligns visual regions with text prompts [15] for Visual Grounding. For each image, the product type extracted from the product metadata (e.g., "*dress*," "*backpack*") was used as the text prompt to grounding dino to generate candidate boxes along with confidence scores. The top-scoring box was selected, and its region was cropped and resized. If no box exceeded a confidence threshold the original image was retained to avoid losing critical context. Visual Grounding's ability to leverage semantic text prompts ensures precise localization of product-centric regions, reducing noise from irrelevant backgrounds (e.g., studio props). To enhance the focus on product-relevant visual elements, we employ the following steps to refine image inputs:

Given an image I, Grounding DINO generates a set of N bounding box proposals:

$$B = \{b_1, b_2, \ldots, b_N\}$$

Each bounding box $b_i \in B$ is associated with a confidence score s_i:

$$s_i = \frac{\exp(\phi_{\text{image}}(v_i) \cdot \phi_{\text{text}}(P)/\tau_{\text{DINO}})}{\sum_{j=1}^{N} \exp(\phi_{\text{image}}(v_j) \cdot \phi_{\text{text}}(P)/\tau_{\text{DINO}})}$$

where $\phi_{\text{image}}(v_i)$ and $\phi_{\text{text}}(P)$ represent the Grounding DINO's encoders for the image region v_i and text prompt P, τ_{DINO} is the temperature parameter, and s_i represents the probability of b_i being the most relevant region. The highest-confidence bounding box b^* is selected using:

$$i^* = \arg \max_{i \in \{1,\ldots,N\}} s_i$$

If the confidence score of b_{i^*} is below a pre-defined threshold τ_{thresh}, the full image is retained:

$$I_{\text{crop}} = \begin{cases} \text{Crop}(I, b_{i^*}), & \text{if } s_{i^*} \geq \tau_{\text{thresh}} \\ I, & \text{otherwise} \end{cases}$$

where $\text{Crop}(I, b_{i^*})$ extracts the product-centered region based on the selected bounding box, and I_{crop} is the final refined image input. Once the refined image I_{crop} is obtained, it is passed through the CLIP vision encoder $\phi_{\text{CLIP-image}}$ to obtain its feature embedding:

$$v = \frac{\phi_{\text{CLIP-image}}(I_{\text{crop}})}{\|\phi_{\text{CLIP-image}}(I_{\text{crop}})\|}$$

where v is the normalized image embedding. By leveraging Visual Grounding for region refinement, we ensure that the extracted embeddings capture fine-grained product attributes, leading to improved alignment in multimodal retrieval.

3.2 LLM-driven Textual Query Synthesis

To improve textual representations for multimodal retrieval, we introduce an **LLM-driven text refinement process**. This process enhances product descriptions by generating structured and semantically rich queries that align better with visual features. The

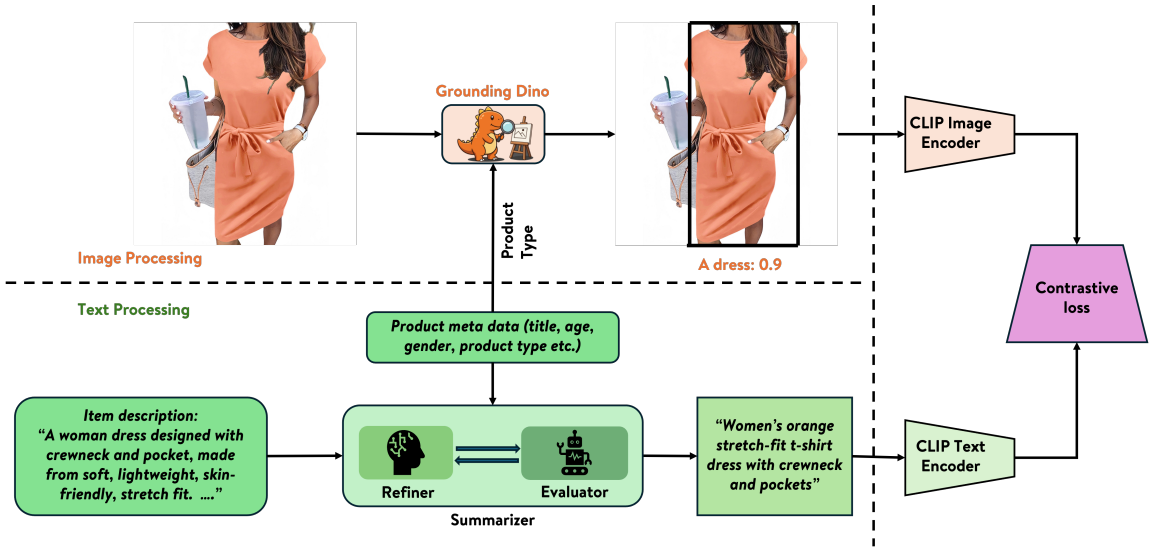


Figure 2: VL-CLIP model architecture

approach consists of three main components: *Summarization, Evaluation, and Refinement.*

Given a raw textual input consisting of both structured and unstructured product information, we first construct an initial concatenated metadata representation as t_{concat}.

$$t_{\text{concat}} = [t_p \parallel t_g \parallel t_{\text{raw}} \parallel t_{\text{in-context}}]$$

where t_p denotes the *product type* (e.g., "*t-shirt,*" "*handbag*"), t_g represents *age and gender attributes* (when applicable), t_{raw} represents the original *product title and description* and $t_{\text{in-context}}$ contains few-shot examples curated to guide the LLM's behavior in ambiguous cases. This concatenated information is summarized by an LLM-based summarizer to form the initial query q^{init}.

$$q^{\text{init}} = \text{Summarizer}(t_{\text{concat}})$$

Given the recent advances have demonstrated strong few-shot capabilities of LLMs [2], we leverage curated set of few-shot examples, specifically designed to address scenarios where LLM exhibits misalignment in $t_{\text{in-context}}$. This allows us to reinforce the desired behavior and improve performance, while maintaining model generality.

Next, we iteratively refine this initial query into a structured, concise, and visually relevant query using two specialized LLM-based modules: an Evaluator and a Refiner.

Let $\text{Evaluator}(q, t_{\text{concat}})$ be an LLM-based function assessing query q's quality against the concatenated input text t_{concat} based on these criteria:

(1) **Attribute Consistency**: Ensures the query reflects attributes present in the input. For example, if q specifies color as red, this criterion evaluates whether the t_{concat} contains a color attribute and that it is indeed red.
(2) **Conciseness**: Limits length of query to *10–20 words* while preserving the meaning.
(3) **Alignment with Visual Data**: Retains only visually discernible attributes. For example, if the t_{concat} mentions a t-shirt is "striped and quick-drying", this criterion ensures we retain only "striped" since it's visually discernible, while excluding "quick-drying" as a non-visual functional attribute.

The Evaluator outputs either a refinement suggestion or a special token <STOP> when no further improvements are necessary. Let $\text{Refiner}(q, e)$ be an LLM-based function that generates a refined query using the current query q and feedback e from the Evaluator. We denote the Evaluator's output and refined query at iteration i as e^i and q^i, respectively.

Starting with q^{init} as q^0 here, at each iteration i $(1 \le i \le i_{\max})$, the Evaluator first assesses the query from the previous iteration q^{i-1} and provides feedback $e^i = \text{Evaluator}(q^{i-1}, t_{\text{concat}})$. If the Evaluator indicates that no further improvement is necessary by returning <STOP>, the iterative refinement process terminates, and the query q^{i-1} is accepted as final. Otherwise, the Refiner function uses the Evaluator's feedback to generate an improved query for the next iteration $q^i = \text{Refiner}(q^{i-1}, e^i, t_{\text{concat}})$. We empirically set $i_{\max} = 5$ as this provides sufficient iterations for convergence while maintaining computational efficiency.

After the iterative refinement concludes, we obtain the final refined query, denoted as q^{final}. This query is then embedded into a semantic space suitable for multimodal retrieval by a **text encoder** ϕ_T, producing a normalized embedding vector t:

$$t = \frac{\phi_{\text{CLIP-text}}(q^{\text{final}})}{\|\phi_{\text{CLIP-text}}(q^{\text{final}})\|}$$

where t represents the **normalized textual embedding** used for matching against the **image embeddings** in the retrieval model. By employing this **LLM-driven synthesis method**, the textual representations become **more structured, visually aligned, and domain-adapted**, ultimately enhancing the performance of the

multimodal retrieval system. This iterative loop illustrated in Figure 3, echoing the self-reflection and self-correction mechanisms, allows the model to autonomously improve its output.

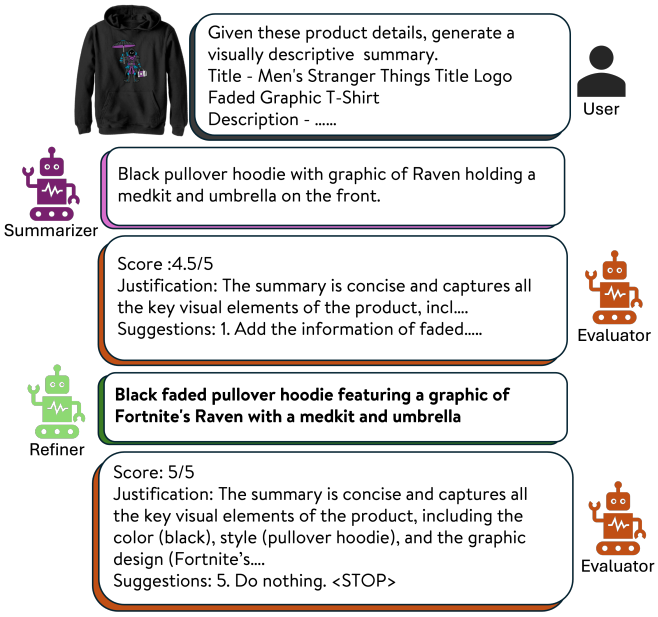


Figure 3: Visualization of product summary generator

The prompts used for the Summarizer, Evaluator, and Refiner are provided in Appendix C.1.

3.3 Contrastive Fine-tuning of CLIP

We align image and text embeddings in a shared semantic space to fine-tune CLIP, overcoming general-purpose model limitations. We employ a symmetric contrastive loss function, maximizing similarity between matched image-text pairs while minimizing it for mismatches. This ensures robust alignment across modalities. A fine-tuned ViT-B/32 processes cropped images, while a transformer-based text encoder refines LLM-augmented queries. Both produce 512-dimensional embeddings optimized for e-commerce-specific retrieval tasks. Training involves multiple epochs, leveraging domain-specific augmentations to achieve higher precision in retrieval and classification tasks. This introduces a systematic framework for fine-tuning the CLIP model to achieve robust image-text alignment in e-commerce applications. The symmetric InfoNCE-style loss maximizes similarity for matched pairs and minimizes it for negatives:

$$\mathcal{L}_{\text{CLIP}} = -\frac{1}{2N}\sum_{i=1}^{N}\left[\log\frac{e^{v_i\cdot t_i/\tau}}{\sum_{j=1}^{N}e^{v_i\cdot t_j/\tau}} + \log\frac{e^{t_i\cdot v_i/\tau}}{\sum_{j=1}^{N}e^{t_i\cdot v_j/\tau}}\right]$$

where τ is the temperature of contrastive loss. We summarize the step-by-step procedure for the VL-CLIP training in Algorithm 1 in Appendix C.

3.4 Online Deployment and Scalability

In this section, we introduce our pipeline and how we deploy VL-CLIP at scale over tens of millions of shopping items in Walmart's e-commerce platform. The production inference pipeline combines multimodal processing, efficient indexing, and scalable retrieval to provide recommendations for e-commerce applications. In the following, we detail each component, how we scale it, and its role in the system.

3.4.1 Image and text preprocessing. We leverage perceptual hashing (pHash)[30], a technique that generates compact and robust hash representations of images, which generates fingerprints invariant to resizing and compression. Images were hashed using perceptual hashing techniques to identify and remove duplicates, reducing redundancy in the catalog. After de-duplication, images are processed by Visual Grounding to crop product-centric regions. This reduces false positives caused by background variations (e.g., the same dress on different mannequins). Visual Grounding dynamically crops product-centric regions using metadata-derived prompts (e.g., "*handbag*").

3.4.2 Hierarchical Navigable Small World (HNSW) index. Embeddings are indexed using HNSW[19], a graph-based Approximate Nearest Neighbors (ANN) algorithm optimized for high recall and low latency. The hierarchical graph structure allows logarithmic-time search complexity. Metadata (e.g., product type) is fused with cropped images to create a unified dataset. This ensures retrieval accounts for both visual and contextual signals. Instead of computing image embeddings for all images in the catalog, we maintain an image embedding database. Generating embeddings for large-scale e-commerce data items at million level is computationally intensive. To handle this, we distribute the workload across multiple machines, each equipped with a T4 GPU.

3.4.3 Retrieval and pairwise ranking. For a query embedding e, the HNSW index retrieves top-k candidates using cosine similarity. The ANN index was queried to retrieve visually similar items. Efficient index construction and retrieval are crucial for real-time performance. We optimized the process by grouping items based on product type and constructing separate indices for each group.

3.4.4 Scalability. The architecture developed in this work is now fully deployed on Walmart's e-commerce platform, supporting real-time recommendations and multimodal retrieval at scale. The pipeline integrates data preprocessing, embedding generation, and retrieval in a seamless workflow. These optimizations reduce search space and memory usage while preserving quality. pHash improves MRR by +7.2%; product type-based HNSW indexing improves Precision@1 by +9% and reduces latency by +81% compared to IVF indexing. Algorithm 2 in Appendix D shows the inference procedure.

4 Experiments

4.1 Data Preparation

Millions of product images and metadata (e.g., descriptions, titles, attributes) are sourced from an extensive e-commerce catalog. This diverse dataset includes apparel and home goods, ensuring comprehensive representation of categories. Each sample includes product

images, which may be high-quality but could contain distracting elements in the background, such as real-life settings or lifestyle scenes, as well as textual metadata, which consists of structured attributes (product type, gender, age group) and unstructured data (titles, descriptions).

We leverage following pre-processing steps to clean the input data: 1) Image Normalization: Resized the images and normalized using CLIP's preprocessing pipeline $I_{norm} = \frac{I_{resized} - \mu}{\sigma}$, where μ and σ are channel-wise mean and standard deviation values. 2) Text Sanitization: Removed HTML tags, special characters, and redundant keywords from metadata. Descriptive keywords are retained, while noise (e.g., "*free shipping*") is excluded, yielding semantically rich inputs. 3) Category Balancing: Stratified sampling ensured proportional representation of product types to mitigate bias that can skew model predictions toward overrepresented categories.

We fine-tune the VL-CLIP model using 7 million products from the fashion and home categories of *Walmart.com* using model architecture described in Figure 2. We evaluated our model on a dataset containing fashion and home items. To ensure variety, we sampled items equally across different product types—such as T-shirts, dresses, and coffee tables—resulting in 10 product types for fashion and 7 for home, for a total of 17 product types. In total, we obtained 10,000 samples for fashion category and 10,000 samples for home category for evaluation.

4.2 Evaluation Metrics

The performance of VL-CLIP is compared with existing methods including CLIP [23], GCL [33], and FashionCLIP [5] on multi-modal retrieval task on Walmart data. CLIP is a foundational model that learns joint representations from large-scale image–text pairs through contrastive learning [23]. GCL is a generalization of contrastive learning framework that incorporates ranking information alongside multiple input fields containing image-text pairs and queries [33]. FashionCLIP is a specialized adaptation of the CLIP paradigm designed for the fashion domain, leveraging fine-grained annotations and domain-specific features [5].

We measure retrieval performance using two standard metrics:

- **HITS@k:** This metric reports the fraction of queries for which the correct item is among the top k results in the ranked list. Formally, for N queries, each query i has a ground-truth correct item c_i. After ranking all items according to a similarity score, let $\text{rank}(c_i)$ be the position of c_i. Then $\text{HITS@k} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}(\text{rank}(c_i) \leq k)$, where $\mathbf{1}(\cdot)$ is an indicator function that returns 1 if $\text{rank}(c_i) \leq k$ and 0 otherwise. In our evaluation, we use HITS@5.
- **Mean Reciprocal Rank (MRR):** For a query i, if the correct item c_i is ranked at $\text{rank}(c_i)$, its reciprocal rank is $\text{RR}_i = \frac{1}{\text{rank}(c_i)}$. The MRR is the average of these reciprocal ranks across all N queries, given by $\text{MRR} = \frac{1}{N}\sum_{i=1}^{N} \text{RR}_i$. This metric particularly favors correct items that rank higher in the list.

4.3 Retrieval Results

Table 1 illustrates how CLIP, GCL, FashionCLIP, and our proposed VL-CLIP perform on the Fashion and Home datasets, using the HITS@5 and MRR metrics. CLIP, the baseline pre-trained vision-language model, shows modest retrieval capability (HITS@5 of 0.3080 on Fashion and 0.2355 on Home), likely because its global embeddings struggle to capture fine-grained product attributes. The multi-modal retrieval task involves identifying the most relevant image from a given set based on a textual description. For example, in a product retrieval scenario, the goal is to match a product description with its corresponding image in a catalog.

GCL improves upon CLIP by integrating fine-grained relevance scores into the contrastive learning process, allowing it to explicitly learn nuanced ranking signals rather than binary relevance alone, thus achieving higher metrics (HITS@5 of 0.3992 on Fashion and 0.3104 on Home). However, its reliance on ranking information alone does not fully address domain-specific nuances in product images and textual descriptions.

FashionCLIP further improves the performance (HITS@5 of 0.4428 on Fashion and 0.4227 on Home) by applying domain adaptation strategies optimized for fashion. This adaptation allows the model to better encode style and design elements that are particularly relevant for apparel, yet it also provides a notable boost on the Home dataset, indicating that fine-tuning vision-language representations with domain-aware features can generalize beyond the original domain.

VL-CLIP delivers the highest retrieval accuracy and ranking quality across both datasets, as demonstrated by its leading HITS@5 and MRR scores (0.6758 and 0.5252 on Fashion, and 0.6692 and 0.5100 on Home). By integrating local object-level grounding for visual representations and large language model–enriched text embeddings, VL-CLIP captures key product details and resolves ambiguous textual descriptions more effectively than competing methods. The result is a more precise alignment between images and text, which proves especially valuable in e-commerce scenarios where seemingly subtle product attributes and nuanced language can critically impact retrieval success.

Table 1: Multi-modal retrieval performance of different models on Fashion and Home datasets

Model	Fashion		Home	
	HITS@5	MRR	HITS@5	MRR
CLIP	0.3080	0.2387	0.2355	0.1747
GCL	0.3992	0.2952	0.3104	0.2312
FashionCLIP	0.4428	0.3555	0.4227	0.3219
VL-CLIP	**0.6758**	**0.5252**	**0.6692**	**0.5100**

4.4 Ablation Study

To gain deeper insight into the role of each component within the VL-CLIP framework, we perform an ablation study by eliminating essential modules, Visual Grounding and LLM-based query refinement, and assessing how their removal affects retrieval performance.

The results of this ablation analysis are summarized in Table 2. The full VL-CLIP model achieves the highest performance with a

HITS@5 of 0.6758 and an MRR of 0.5252. Removing Visual Grounding results in an average performance drop of 15.34% in HITS@5 and 11.23% in MRR across the Fashion and Home categories, demonstrating the importance of background removal and focusing on the main item in enhancing visual matching. Additionally, removing the LLM-based query refinement step further reduces performance by 7.40% in HITS@5 and 5.32% in MRR when compared to the model already lacking Visual Grounding, indicating that refining text queries improves retrieval accuracy by providing clearer and more precise textual descriptions. This ablation study highlights that both Visual Grounding and LLM-based query enhancement play crucial roles in improving retrieval effectiveness.

Table 2: Ablation study on the contribution of each component for Fashion and Home datasets

Model Variant	Fashion		Home	
	HITS@5	MRR	HITS@5	MRR
VL-CLIP w/o GD, LLM	0.4484	0.3570	0.4418	0.3471
VL-CLIP w/o GD	0.5308	0.4176	0.5075	0.3929
VL-CLIP	**0.6758**	**0.5252**	**0.6692**	**0.5100**

4.5 Zero-shot Classification

In addition to the information retrieval and the ablation test, we also performed a zero-shot classification task. We performed two fashion item attribute classification tasks: neckline classification and pattern classification. For neckline classification, we manually selected 1,000 fashion items, each belonging to one of the following categories: v-neck, crew neck, scoop neck, Henley, mock neck, and boat neck. We use a zero-shot classification approach, where we generate a descriptive text for each class (e.g., "*a T-shirt with a scoop neckline*") and pass it through a text encoder. The classification is then performed by comparing the image embedding with these text embeddings to find the closest match, which determines the predicted class. Similarly, for pattern classification, we apply the same zero-shot approach using the following categories: "*solid,*" "*cartoon character,*" "*heart symbol,*" and "*floral print*".

Table 3 presents the model accuracy for both classification tasks. VL-CLIP consistently outperforms other models, making it the most reliable choice for fashion attribute zero-shot classification. Its superior performance is due to Visual Grounding's ability to remove noise and the LLM-refined queries, which enhance the quality of text-image alignment.

4.6 VLM-Agent Evaluation

Since the alignment of text and image information is very subjective, we employ a VLM agent for evaluation. Our evaluation consists of two retrieval tasks: query-based retrieval and similar item recommendation. The query-based retrieval specifically targets fine-detailed product attributes to ensure accurate retrieval of nuanced product characteristics. For example, "*Teal floral print blouse*" is looking for items that match color and pattern characteristics; "*Beige V-neck short-sleeve T-shirt*" is looking for color, neckline and sleeve characteristics. For query-based evaluation, the retrieved images corresponding to each query are individually paired with the query and passed to a VLM. The VLM model is asked to assess whether the provided image accurately matches the given query, producing a binary output of 0 (no match) or 1 (match). Similarly, for similar item evaluation, the retrieved images are individually paired with the anchor image, and the VLM is asked to assess whether the two images match in terms of their visual characteristics. We assess the effectiveness of our approach using an VLM-as-judge evaluation framework. More details on this process for automated query generation and VLM-evaluation are provided in Appendix E.

Table 3: Zero-shot performance on pattern and neckline classification tasks

Model	Neckline classification accuracy	Pattern classification accuracy
CLIP	0.580	0.144
GCL	0.674	0.785
FashionCLIP	0.881	0.934
VL-CLIP	**0.937**	**0.959**

Table 4 presents the query-based retrieval performance and similar item recommendation performance for *Walmart.com E-Commerce Dataset*. Performance is reported using Precision@1, 3, 5. The results show that our VL-CLIP model perform better than the benchmarks including CLIP, FashionClip, and GCL. Note here the highest values for VL-CLIP appear at Precision@1, gradually decreasing for Precision@3 and Precision@5. This pattern indicates that its top-ranked item is almost always relevant, while subsequent positions, though still relevant, can exhibit slightly lower relevance. In contrast, models like CLIP sometimes show a reversed pattern—with lower Precision@1 than Precision@5—suggesting that their top recommendation is not always the best match, even though they do include relevant items in lower-ranked positions. Examples of both query based and Similar Item (SI) recommendation tasks are provided in Appendix B.

The improvements can be attributed to the complementary roles of Visual Grounding and LLM in refining the retrieval process. Visual Grounding helps the model focus on the main item within the image, filtering out background distractions, and ensuring that fine-detailed product attributes are emphasized. Meanwhile, LLM enhances the quality of search queries by making them more structured and aligned with real-world user intent. Together, these enhancements enable for more accurate retrieval of products that match specific attribute-based queries.

4.7 Computation Efficiency

VL-CLIP is fine-tuned on millions of products from the fashion and home categories of *Walmart.com*. Stratified sampling method is applied to ensure proportional representation of diverse set of product types (more than 500 product types). VL-CLIP achieved robust performance on the e-commerce retrieval task over 6 epochs before early stopping (see Figure 4). The model demonstrated strong alignment between visual and textual embeddings, evidenced by a steady reduction in the contrastive loss for the validation set from 0.38

Table 4: VLM-evaluation results of query-based retrieval and similar item recommendation

Query-based retrieval			
Model	**Precision@1**	**Precision@3**	**Precision@5**
CLIP	0.3800	0.4500	0.4620
FashionCLIP	0.5900	0.6833	0.7100
GCL	0.4500	0.4800	0.4880
VL-CLIP	**0.8586**	**0.7710**	**0.7515**
Similar item recommendation			
Model	**Precision@1**	**Precision@3**	**Precision@5**
CLIP	0.9719	0.9046	0.8680
FashionCLIP	0.9813	0.9582	0.9439
GCL	0.9813	0.9576	0.9349
VL-CLIP	**0.9925**	**0.9838**	**0.9783**

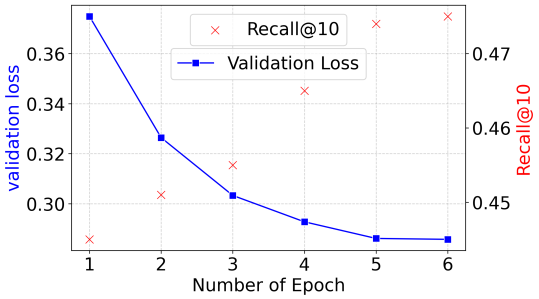


Figure 4: The validation loss and recall@10 over epochs

to 0.28. Retrieval performance, measured by *Recall*@10 indicating that the model effectively identified relevant items in the top-10 results for 47% of queries. Prolonged training beyond this point led to a marginal decline in Recall@10, suggesting overfitting to noisy pairs or saturation in learning capacity. This underscored the importance of early stopping, with epoch 6 representing the optimal checkpoint for deployment. These results validate the effectiveness of our pipeline—combining Visual Grounding, LLM, and contrastive loss—for scalable e-commerce recommendation systems.

4.8 Cross-Domain Generalization

To assess the generalizability of VL-CLIP, we conduct zero-shot evaluations on a public Google Shopping dataset[1]. This data set spans various e-Commerce categories and provides a robust benchmark for testing the model's ability to transfer knowledge to unseen domains without additional fine-tuning. It is specifically designed for training and benchmarking multi-modal retrieval models in fine-grained ranking tasks. As shown in Table 5 and Table 6, VL-CLIP consistently outperforms other models when applied to this new dataset.

[1]https://github.com/marqo-ai/GCL

We further evaluate zero-shot performance on the *Art* and *Toys* categories from *Walmart.com*, where VL-CLIP again achieves superior results compared to other models. These findings highlight the model's strong transferability to novel product domains (see Appendix F).

Table 5: Multi-modal retrieval performance of different models on Google Shopping dataset

Model	**HITS@5**	**MRR**
CLIP	0.2419	0.1714
FashionCLIP	0.4495	0.3075
GCL	0.6270	0.4283
VL-CLIP	**0.6644**	**0.4936**

Table 6: VLM-evaluation results of query-based retrieval and similar item recommendation on Google Shopping dataset

Query-based retrieval			
Model	**Precision@1**	**Precision@3**	**Precision@5**
CLIP	0.3935	0.4258	0.4167
FashionCLIP	0.7032	0.7182	0.7238
GCL	0.4193	0.4107	0.4129
VL-CLIP	**0.8452**	**0.8215**	**0.7896**
Similar item recommendation			
Model	**Precision@1**	**Precision@3**	**Precision@5**
CLIP	0.6161	0.5684	0.5423
FashionCLIP	0.8298	0.7980	0.7796
GCL	0.7759	0.7434	0.7141
VL-CLIP	**0.9294**	**0.9073**	**0.8950**

4.9 Online A/B Test

To validate the effectiveness of VL-CLIP model, we conduct a large-scale A/B test on one of the top two e-commerce platforms in US. The experiment compared the performance of our VL-CLIP with the deployed baseline model. The test lasted four weeks and included millions of user interactions in various product categories. The following key metrics are evaluated in the AB test: Click-Through Rate (CTR), the proportion of users who clicked on recommended products after viewing them; Add-to-Cart Rate (ATC), the percentage of users who added a recommended product to their cart; Gross Merchandise Value (GMV), the total sales revenue generated by the recommendations.

Table 7 highlights the relative improvements of our system compared to the baseline model. Online A/B tests validated the effectiveness of VL-CLIP, revealing an 18.6% increase in CTR, a 15.5% increase in ATC rate, and a 4% boost in GMV, underscoring the VL-CLIP's practical efficacy. These results highlight the performance of VL-CLIP in understanding user intent and aligning recommendations with user preferences.



Figure 5: Examples of similar item recommendations for fashion products based on VL-CLIP.

Table 7: Online A/B test results

Performance metric	Relative improvement
CTR (%)	18.6%
ATC(%)	15.5%
GMV(%)	4.0%

We show some case studies in Figure 5. The first column is the anchor item, the rest are top five recommended items based on VL-CLIP. In Figure 5(a), the anchor item is a green floral midi dress. VL-CLIP retrieves similar style dresses, capturing variations in pattern and length while maintaining the overall aesthetic. Figure 5(b) item is a black wrap-style dress with long sleeves. VL-CLIP recommends items with similar sleeve lengths and structured silhouettes, focusing on both color and style. Figure 5(c), (d), and (e) demonstrate the strong fashion understanding capability of VL-CLIP. For further case studies, please refer to Figure 6-8 in Appendix B.

5 Conclusion and Future Work

In this work, we addressed critical challenges in multimodal representation learning for e-commerce by introducing VL-CLIP, a novel framework that integrates Visual Grounding for visual representation enhancement and LLM-augmented text embeddings for semantic enrichment. Through extensive experiments on large scale e-commerce datasets, VL-CLIP demonstrated superior performance over state-of-the-art baselines. Specifically, HITS@5 improved by 184.16% on Home dataset and by 119.42% on the Fashion dataset. Furthermore, LLM evaluation results indicate a 62.66% increase for query-based retrieval and a 12.71% improvement in similar item recommendations. Online A/B tests further validated the effectiveness of VL-CLIP, revealing an 18.6% increase in CTR, a 15.5% increase in ATC rate, and a 4% boost in GMV, underscoring the VL-CLIP's practical efficacy. Deploying VL-CLIP on *Walmart.com* highlighted its scalability and real-world impact. The framework's hierarchical indexing and distributed computation pipeline efficiently processed millions of catalog items.

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